

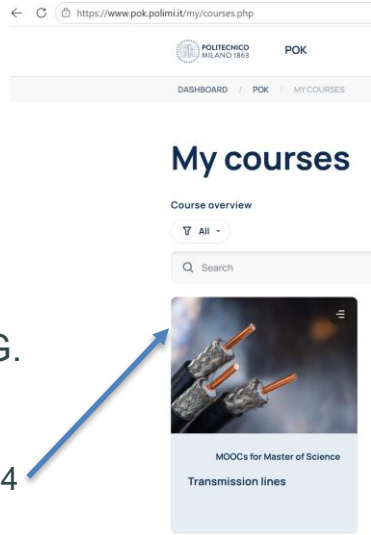
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APPLIED ELECTROMAGNETICS @POLIMI

RF Systems

Guided Propagation (Transmission Lines)

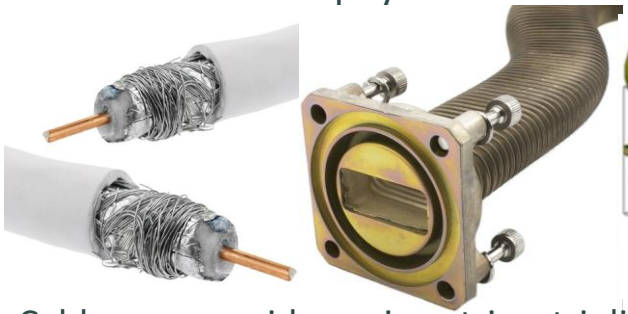
Recommended: introduction by Prof. G. G. Gentili on Polimi Open Knowledge "Transmission Lines" (tot 1h):
https://www.pok.polimi.it/course/view.php?id=194

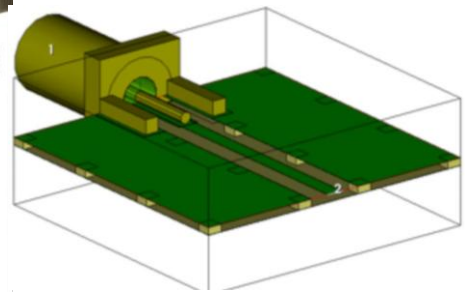


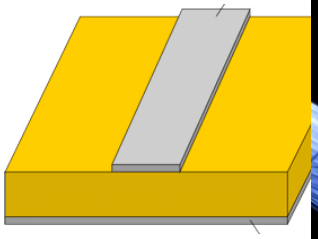
Basic Concept of Transmission Line

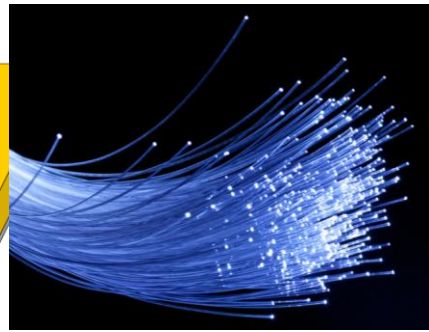
Antenna: interface between free space propagation and guided propagation

Transmission line: physical structure where guided propagation occurs









Cables, waveguides, microstrip, stripline, coplanar, slotline, optical fibers…

Fields are confined into a precise region determined by the cross section (shapes and materials) and boundary conditions: power density (Poynting vector)

Waves can propagate along the line, generally following its path

Lossless lines: power density is preserved during propagation (no attenuation)

Physical lines (lossy): a fraction of power is dissipated by conductor and dielec. losses →attenuation (dB/m)

EM Fields inside a Transmission Line

Analysis: given materials and shape of the cross-section, determine the "shape" of electric and magnetic fields

→ Solve Maxwell equations (or simulate to obtain an approximate solution)

An arbitrary field (vector phasor at ω) propagating toward +z in a tx line is usually expressed as:

$$\vec{E}_{tot}(x,y,z) = \sum \vec{E}_i(x,y) e^{-j\beta_i z} e^{-\alpha_i z} \qquad \vec{H}_{tot}(x,y,z) = \sum \vec{H}_i(x,y)\, e^{-j\beta_i z} e^{-\alpha_i z}$$

where $E_i(x,y)$, $H_i(x,y)$, β_i, α_i are the *modes* of the structure

Each mode (e.g. TEM, TE10, TM11…) has

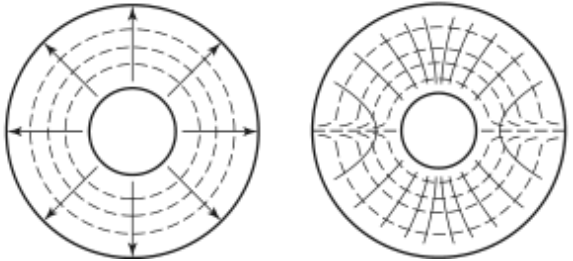
- its own propagation constant $\gamma_i = \alpha_i + j\beta_i$
- Its own wave impedance Z_W (ratio between E_i and H_i)

and all depend on ω

Modes can propagate only if $\beta > 0$

Some modes have $\beta > 0$ only above a given frequency, called *cutoff frequency* of the mode

α_i: attenuation constant (in Np/m); **1Np=8.686dB** → Easy conversion from dB/m to Np/m

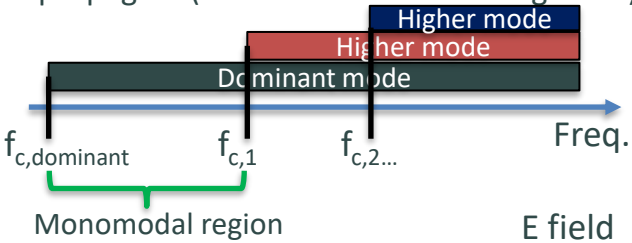


Examples of 2 modes in a coaxial cable

Classification of modes

Dominant mode: the mode with the lowest cutoff frequency (normally used in the practice for that line)

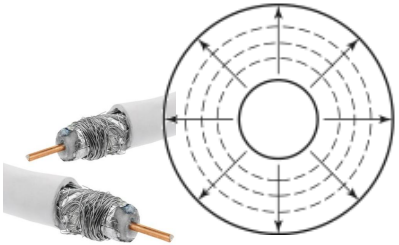
Monomodal region: frequency range where only the dominant mode can propagate (all other modes have higher fc)

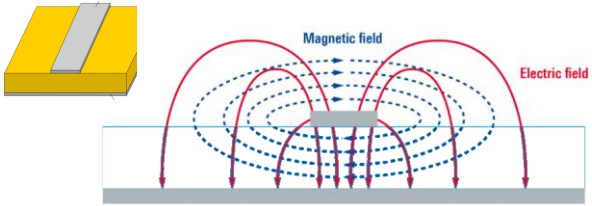


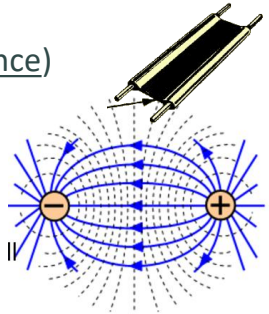


If the cutoff frequency of the dominant mode is zero ($f_{c,dominant}$ =0):

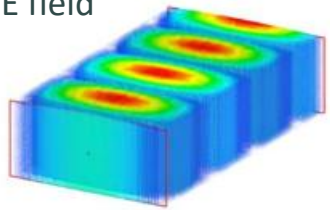
- propagation is possible from ~DC
- the mode is called principal (or Transverse Electro-Magnetic, TEM)
- To exist, TEM requires at least two separate conductors.
- Voltage and currents can be defined univocally and used instead of E and H fields
- For a TEM mode the propagation constant β depends linearly on ω
- wave impedance Z_W is related to a circuit impedance Zc (characteristic impedance)



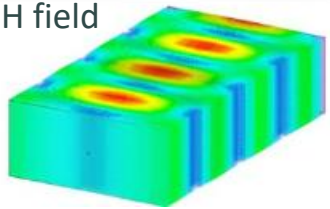




E field

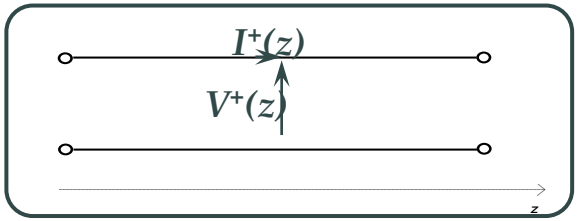


H field



Phase velocity of voltage and current waves in a TEM line

An infinitely long transmission line where one mode is propagating can be schematically represented by two wires of arbitrary length in the *z* direction.



A voltage wave travelling toward +z is:

$$V^{+}(z,t) = Re\left(V_0^{+} e^{-j\beta z} e^{-\alpha z}\, e^{j\omega t}\right) = |V_0^{+}| \mathrm{e}^{-\alpha z} \cos(\omega t - \beta z + \angle V_0^{+})$$

Phasor: $V^{+}(z) = V_0^{+} e^{-j\beta z} e^{-\alpha z}$ and current $\mathrm{I}^{+}(\mathrm{z}) = \frac{V_0^{+}}{Z_C} e^{-j\beta z} e^{-\alpha z}$

In t+Δt, we can find the variation Δz of z for which the phase of the wave remains constant:

$$\omega \cdot \Delta t - \beta \cdot \Delta z = 0 \;\Rightarrow \beta = \frac{\omega}{\frac{\Delta z}{\Delta t}} = \frac{\omega}{v}$$

(can be ≠ from wavelength in vacuum)

where *v* is called *phase velocity* of the wave: Phase constant: $\beta = \frac{\omega}{v}$ Wavelength in the line: $\lambda = \frac{2\pi}{\beta} = \frac{v}{f}$

A voltage wave can also propagate toward -*z*: $V^{-}(z,t) = Re\left(V_0^{-} e^{j\beta z} e^{\alpha z}\, e^{j\omega t}\right) = |V_0^{-}| \mathrm{e}^{\alpha z} \cos(\omega t + \beta z + \angle V_0^{-})$

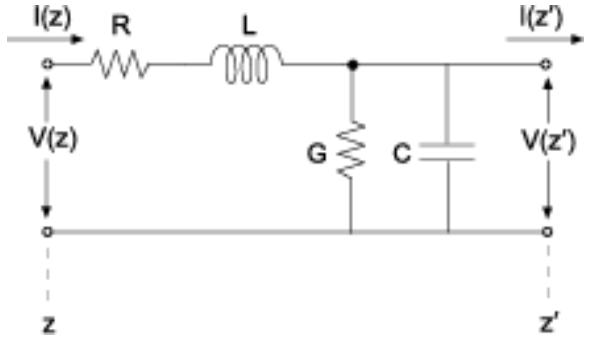
And corresponding phasor $\mathrm{V}^{-}(\mathrm{z}) = V_0^{-} e^{j\beta z} e^{\alpha z}$ and current $\mathrm{I}^{-}(\mathrm{z}) = -\frac{V_0^{-}}{Z_C} e^{j\beta z} e^{\alpha z}$

Primary and secondary parameters of a Line

Z_C (Ω)and v (m/s) and α (dB/m or Np/m): "**secondary parameters**" of the line

Instead, L, C, G, R are "**primary parameters**"

- L: inductance per unit length (H/m)
- C: capacitance per unit length (F/m)
- G: conductance per unit length (determined by dielectric losses) (S/m)
- R: resistance per unit length (determined by conductor losses) (Ω/m)
- We assume G<<ωC, R<<ωL (low-loss line)



$$Z_c = \sqrt{\frac{L}{C}} \in Real, \qquad v = \frac{1}{\sqrt{L \cdot C}} \qquad \beta = \frac{\omega}{v} \qquad \alpha = \frac{GZ_C}{2} + \frac{R}{2Z_C}$$

Datasheets usually provide: α in dB/m (→ needs conversion to Np/m), Zc and sometimes C or β at a given frequency (in °/m)

Attenuation in transmission lines: losses

Power along a line (positive z direction): $P^+(z) = \frac{1}{2} Re\{V^+(z) I^{+*}(z)\} = \frac{1}{2}\frac{|v^+(z)|^2}{Z_c} = \frac{1}{2}\frac{|v_0^+|^2}{Z_c} e^{-2\alpha z}$

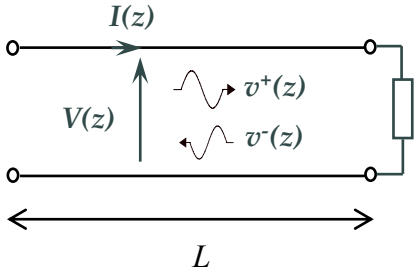
α in these computations is in its natural unit, Np/m

The power incident on a load $Z_L=Z_C$ is then: $P^+(L) = P^+(0)e^{-2\alpha L}$

The difference is the power dissipated by losses in the line

$P_{diss} = P^+(0) - P^+(L) = P^+(0)(1 - e^{-2\alpha L})$

P+(0): Power incident on the Tx Line



P+(L): Power incident on the load

Power dissipated per unit length: $P_{diss\,p.u.l.} = \lim_{L\to 0} \frac{P_{diss}}{L} = \lim_{L\to 0} \frac{1-(1-2\alpha L)}{L} = 2\alpha$

Providing 1kW to a L=5m line with α=0.1dB/m, terminated on a matched load,
α=0.0115Np/m: P(L)=1kW · 0.891=891W → Dissipated by the line: 109W

1 Np=8.686 dB (=20Log(*e*))

Easier in dB: $P(L)_{dBW} = P^+(0)_{dBW} + 10\text{Log}(e^{-2\alpha_{Np/m}L}) = P^+(0)_{dBW} - L\cdot\alpha_{Np/m}\,20\text{Log}(e) = P^+(0)_{dBW} - L\cdot\alpha_{Np/m}\,8.686dB/Np$

$\rightarrow Atten_{dB} = \alpha_{dB/m} L;\rightarrow P(L)_{dBW} = P(0)_{dBW} - Atten_{dB}$ =30dBW-0.1dB/m · 5m=29.5dBW

For noise, a lossy TX line is a resistive attenuator at its physical temperature: $T_{eq} = T_{phys}(10^{Atten_{dB}/10} - 1)$

End of topics for Midterm!

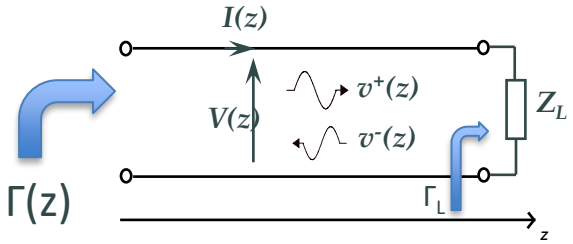
(What we have seen in lectures so far is included. About tx lines: just the basics, e.g. attenuation and primary/secondary parameters)

(What follows is not part of the midterm)

Transmission line of finite length

Physical length L corresponds to an electrical length ϕ at a given frequency *f*: ϕ=βL (where β=ω/v)

In lines with finite length L (terminated by a passive impedance Z_L), two waves are normally present, propagating in opposite direction and having different amplitude:



$$V(z) = v^+(z) + v^-(z) = V_0^+ e^{-j\beta z} e^{-\alpha z} + V_0^- e^{j\beta z} e^{\alpha z}$$

$$I(z) = i^+(z) + i^-(z) = \frac{V_0^+}{Z_C} e^{-j\beta z} e^{-\alpha z} - \frac{V_0^-}{Z_C} e^{j\beta z} e^{\alpha z}$$

v^+, i^+: Incident wave

v^-, i^-: Reflected wave

The ratio between the reflected and incident voltage wave amplitudes is called reflection coefficient

Measuring the position as distance l away from the load (l=L-z):

$$\Gamma(z) = \frac{v^-(z)}{v^+(z)} = \frac{v^-(L-l)}{v^+(L-l)} = \frac{V_0^- e^{j\beta L} e^{-j\beta l} e^{\alpha L} e^{-\alpha l}}{V_0^+ e^{-j\beta L} e^{j\beta l} e^{-\alpha L} e^{\alpha l}} = \frac{v^-(L)}{v^+(L)} e^{-2j\beta l} e^{-2\alpha l} \rightarrow \boxed{\Gamma(z) = \Gamma_L e^{-2j\beta l} e^{-2\alpha l}}$$

→ The observed refl. coeff. varies along the line

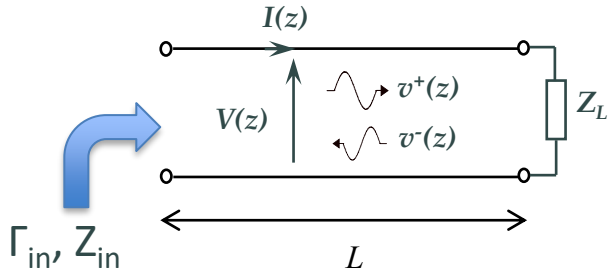
The total voltage can be computed as: $V(z) = v^+(z)\left(1 + \frac{v^-(z)}{v^+(z)}\right) = v^+(z)\left(1 + \Gamma(z)\right)$

Considerations on the reflection coefficient (Z_C real)

In a lossless line (α=0) |Γ(z)| is constant.

$$\Gamma_{in}=\Gamma_L e^{-2j\beta L}$$

If the length is changed by λ/2, Γ_{in} does not change!



with Γ_L given by the mismatch between load and line: $\Gamma_L = \frac{Z_L - Z_C}{Z_L + Z_C}$

When Γ_L=0, no reflected wave exists (the load is matched to the line)

→ Then Γ (z)=0 → Z(z)=V(z)/I(z)=Z_C for all z.

The impedance observed from one end of a line looking toward the other end is:

$$Z_{in} = \frac{V(0)}{I(0)} = Z_c \frac{V^+(0)+V^-(0)}{V^+(0)-V^-(0)} \rightarrow Z_{in} = Z_c \frac{1+\Gamma_{in}}{1-\Gamma_{in}}$$

Some computations are easier using Γ, some using Z, some using Y

→Convert between these (done often!)

Normalized impedance is also used a lot:

Z_n=Z/Z_C (often denoted with lowercase letter *z* as well)

$$Z_n = \frac{Z}{Z_c} = \frac{1+\Gamma}{1-\Gamma} \qquad \Gamma = \frac{Z_n - 1}{Z_n + 1}$$

Γ, Z and Y are equivalent representations at a given point on the line

$$Y_n = \frac{1}{Z_n} = \frac{Z_C}{Z} = \frac{1-\Gamma}{1+\Gamma} \qquad \Gamma = \frac{1-Y_n}{1+Y_n}$$

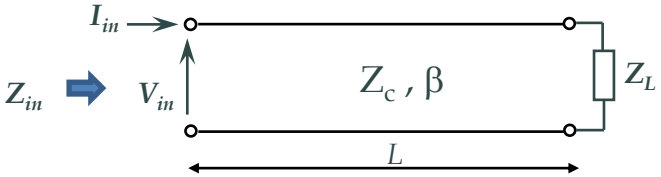
Impedance as a function of the load in a lossless line

We define the **Electrical Length ϕ** of a line with length L at the operating frequency f: $\phi=\beta L=2\pi L/\lambda$

For instance: if $\phi=360°=2\pi$ rad , the physical length is equal to wavelength ($L=\lambda$)

if $\phi=180°=\pi$ rad, the physical length is half a wavelength ($L=\lambda/2$)

if $\phi=90°=\pi/2$ rad, the physical length is a quarter of a wavelength ($L=\lambda/4$)

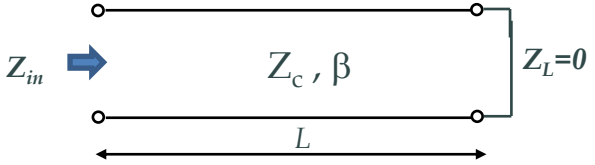


$$Z_{in}=\frac{V_{in}}{I_{in}}=Z_C\,\frac{Z_L+jZ_C\tan\phi}{Z_C+jZ_L\tan\phi}$$

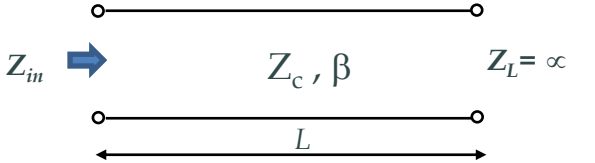
Impedance observed from one end of a line, looking toward the other end, terminated on impedance Z_L

Particular cases (stubs):

Z_L= 0 (short-circuited) ⇒ $$Z_{in}=\frac{V_{in}}{I_{in}}=jZ_c\tan(\phi)$$



Z_L= ∝ (open-circuited) ⇒ $$Z_{in}=\frac{V_{in}}{I_{in}}=-jZ_c\cot(\phi)$$



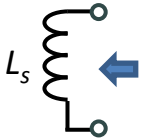
Stub as a circuit component

RECALL SOME NAMES: "Admittance (Y)"=1/"Impedance (Z)"

"Impedance (Z)"="Resistance (R)" + j"Reactance (X)" **"Admittance (Y)"="Conductance (G)" + j"Susceptance (B)"**

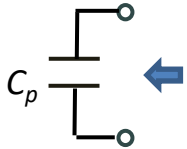
At RF frequency: lumped inductors and capacitors do not really work (lossy, distributed effects…)
But "stubs" can realize the same reactance (susceptance) at a single frequency f_0:

Positive reactance at ω_0

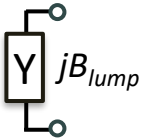


$X_{lump} = \omega_0 L_s$

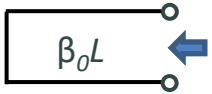




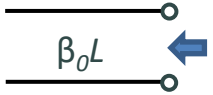
$B_{lump} = \omega_0 C_p$



Positive susceptance at ω_0



$X_{dist} = Z_c \tan(\phi)$



$B_{dist} = Y_c \tan(\phi)$ $with\ Y_c = 1/Z_c$

Impose $X_{dist}=X_{lump}$: $Z_c = \frac{X_{lump}}{\tan(\phi)} = \frac{\omega_0 L_s}{\tan(\phi)}$ L<λ/4 Impose $B_{dist}=B_{lump}$: $Y_c = \frac{B_{lump}}{\tan(\phi)} = \frac{\omega_0 C_p}{\tan(\phi)}$

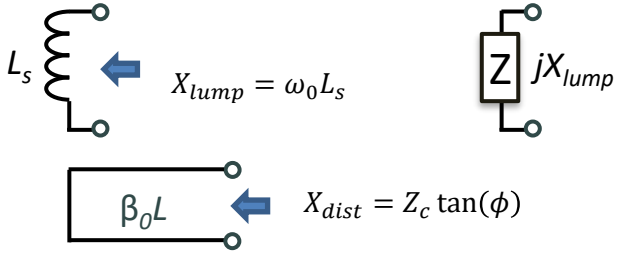
From X_{lump} or B_{lump} we can compute ϕ or Zc at the desired frequency f_0!

→Once ϕ is known, the physical length is L=ϕ/β, with β computed at the desired frequency f_0 ($\beta=2\pi f_0/v$)

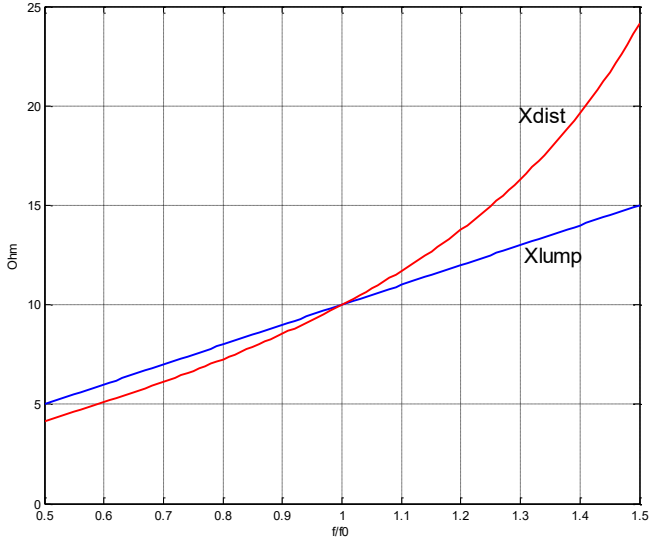
When L>λ/4, other solutions are possible (i.e. obtaining a positive reactance from an open-circuited longer stub)

Example: Lumped element vs Stub

Positive reactance at ω_0



Impose X_{dist}=X_{lump}: $Z_c = \frac{X_{lump}}{\tan(\phi)} = \frac{\omega_0 L_s}{\tan(\phi)}$



Lumped element: $X_{lump}(f_0)= X_{dist}(f_0)= 10\ \Omega$

→ A possible equivalent stub is:

$\Phi=\beta_0 L=45°$, $Zc=10\ \Omega$

The behavior of a lumped inductor in frequency however is: $X_{lump}(f)=10\Omega\cdot(f/f_0)$,

The behavior of a short-circuited stub in frequency instead is: $X_{dist}(f)=10\Omega\cdot\tan(45°\cdot f/f_0)$

The equivalence between a lumped element and a stub is correct ONLY AT f_0!

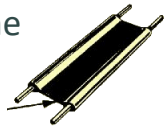
Varying freq., the two behaviors are different (lumped: monotonic; distributed: periodic)

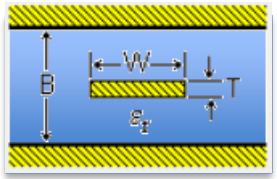
Transmission Lines Classification

☞TEM Lines (Transverse Electric Magnetic)

- Coaxial
- Stripline
- Bifilar line





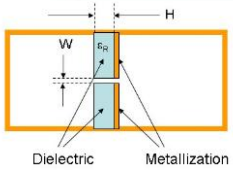


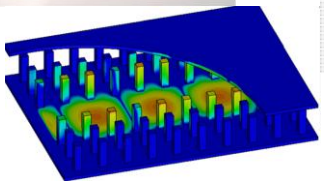
☞non-TEM Lines

- Rectangular waveguide
- Circular waveguide
- Fin-line
- Gap waveguides



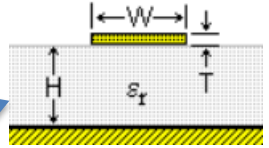


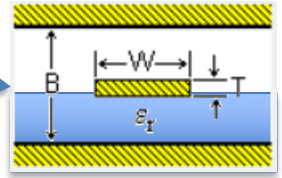


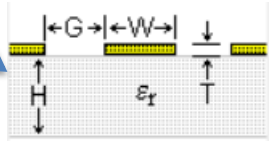


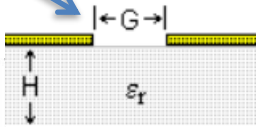
☞Quasi-TEM Lines

- Microstrip
- Suspended Stripline
- Coplanar waveguide (CPW)
- Slotline









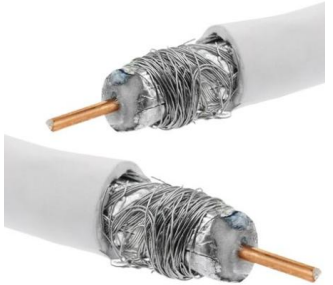
TEM Lines

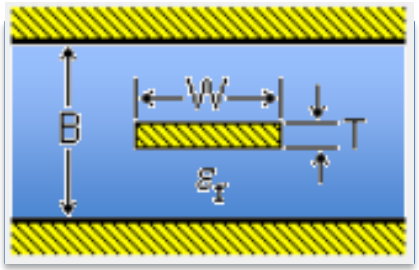
- TEM Lines: constituted by two (at least) independent conductors (a voltage can be applied between them), embedded in a homogeneous medium (ε_r)
- Electric and magnetic fields of the propagating wave are orthogonal to each other and do not have components in the direction of propagation (as in a plane wave)
- A TEM transmission line is characterized by:
  - A univocal and frequency independent characteristic impedance
  - A constant phase velocity lower than or equal to the light velocity c:

$$v = \frac{c}{\sqrt{\epsilon_r}} \qquad \beta = \frac{\omega}{v} = \frac{\omega}{c}\sqrt{\epsilon_r} \qquad \lambda = \frac{2\pi}{\beta} = \frac{c}{f\sqrt{\varepsilon_r}}$$

with ε_r relative dielectric constant (permittivity) of the medium filling the transmission line

[c = velocity of light in vacuum ($3 \cdot 10^8$m/s)]



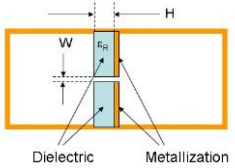


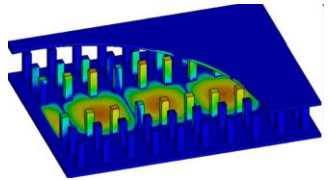
non-TEM Lines

- Lines supporting only modes with cutoff frequency $f_c>0$
- Non-TEM lines are often constituted by a single conductor (e.g. a hollow tube) having a section of arbitrary shape (uniform with *z*).
  Waveguides are non-TEM lines (E and H fields are not orthogonal to *z*)
  Optical fibers are dielectric waveguides, so non-TEM
- The modes in non-TEM lines are generally classified in:
  - **TE modes:** the electric field has no z component
  - **TM modes:** the magnetic field has no z component
  - Hybrid modes: E and H have z component
- The characteristic impedance of a non-TEM mode is not univocally defined
- Voltage and current are not univocally defined.
  The reflection coefficient is instead always unique.



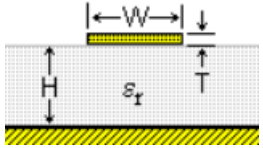


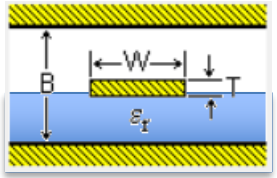


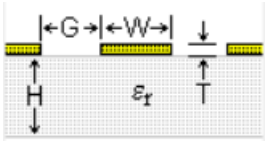


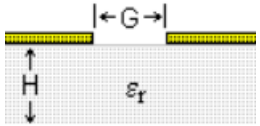
quasi-TEM Lines

- Two (or more) conductors, surrounded by a non-homogeneous medium
- As a consequence, there is at least one component of E or H field in the direction of propag.
- The dominant mode of quasi-TEM lines has cutoff frequency equal to zero (DC) → It can propagate always
- The rigorous study of these lines is complicate; → approx. model: an **equivalent TEM line**
- The equivalent TEM model of a quasi-TEM line differs from an ideal TEM line: characteristic impedance and phase velocity **depend on frequency** (though typically only mildly)
- An effective dielectric constant $\varepsilon_{r,eff}$ is computed (via some models or softwares):





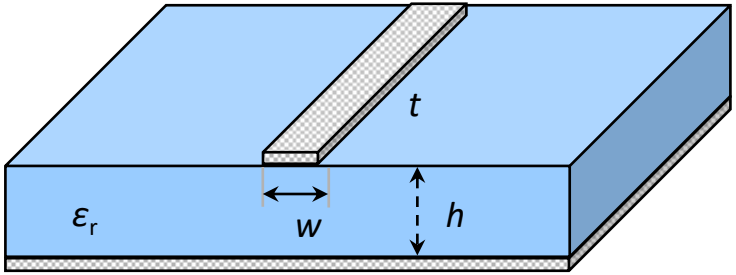




$$Z_c = \frac{Z_{c,\,all\ in\ air}}{\sqrt{\varepsilon_{r,eff}}}, \quad v = \frac{c}{\sqrt{\epsilon_{r,eff}}} \qquad \beta = \frac{\omega}{v} = \frac{\omega}{c}\sqrt{\epsilon_{r,eff}} \qquad \lambda = \frac{2\pi}{\beta} = \frac{c}{f\sqrt{\varepsilon_{r,eff}}}$$

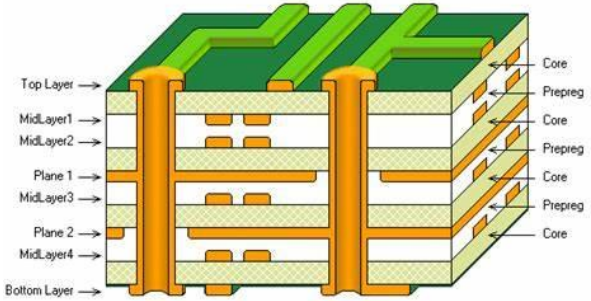
Microstrip

The most important quasi-TEM line in printed circuit boards (PCBs) is the **microstrip**
It belongs to the category of planar structures



Dielectric

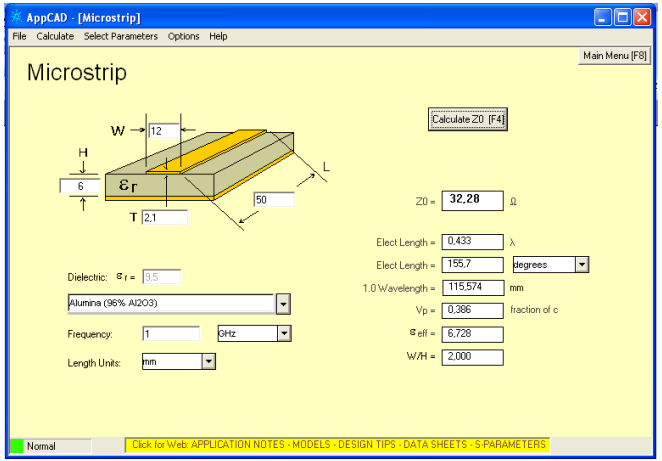
Conductor

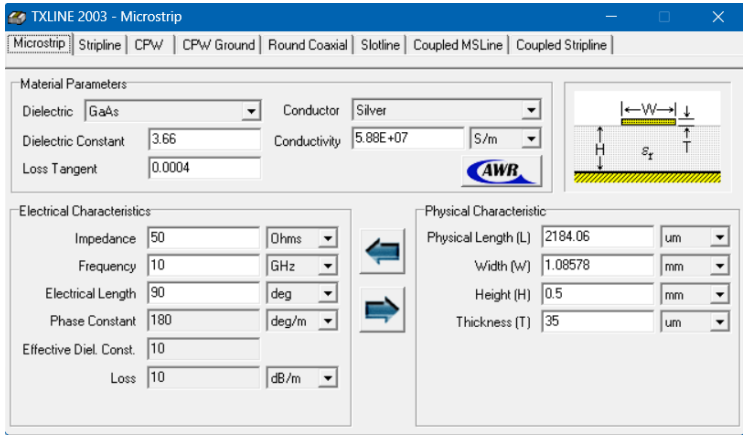


- *h* substrate thickness
- *w* strip width
- ε_r relative dielectric constant of substrate
- *t* metallic thickness

Approximate formulas developed to compute Z_C and $\varepsilon_{r,eff}$

CAD for Microstrip





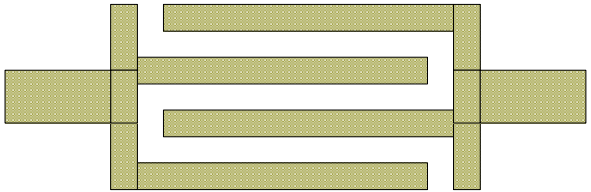
AppCAD, freeware by Broadcom
https://www.broadcom.com/info/wireless/appcad

TX Line, freeware by Cadence https://www.cadence.com/en_US/home/tools/system-analysis/rf-microwave-design/awr-tx-line.html

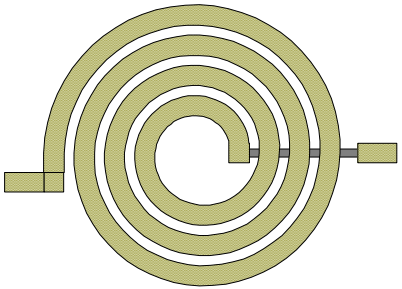
Free online calculators:
https://www.microwaves101.com/calculators/1201-microstrip-calculator
https://emtalk.com/mscalc.php
https://www.pasternack.com/t-calculator-microstrip.aspx

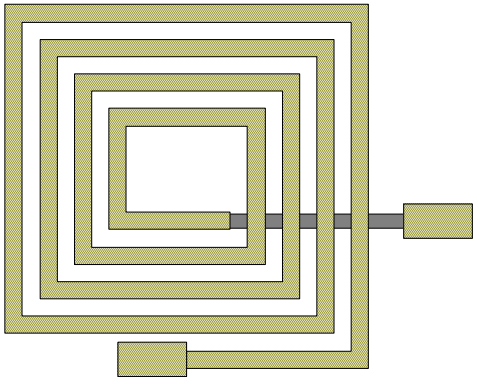
Pseudo-lumped Components in Microstrip

Interdigital capacitor



Spiral Inductors





Microstrip Circuits and Components

